Polynomial-depth learning for ReLU and GELU NNBayes stable under neuron merging

ChatGPT-6 Astra Claude Fable 5 and Opus 5 Kaarel Hänni
Dmitry Vaintrob*

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Abstract

We prove a sample bound polynomial jointly in teacher width and depth for Bayesian ReLU and exact GELU networks with a specified restriction on their prior. The target is a layered NOR, COPY, and NOT circuit of width C, counting a constant-one wire at every stage. One sample event controls posterior margin error simultaneously at every learner width $N \geq C$ for ReLU or $N \geq 2C$ for GELU, and at every mean-field-or-stronger Gaussian scale. The prior retains networks whose clipped predictions change little when neurons are averaged into a limited number of populations. The populations may have arbitrary sizes; their permitted number depends on readout energy and output magnitude. Every stipulated circuit has a positive-mass implementation satisfying this condition. ReLU's positive homogeneity removes population sizes from the compressed computation; a uniformly Lipschitz rescaled activation controls them for GELU. Perturbations may amplify through depth, but only the logarithm of their required parameter precision enters the prior-mass cost. The sample bound is quadratic in teacher width up to logarithms; a coarser bound has literal quadratic width dependence and degree nine in the remaining resources. The population restriction is an assumption on the prior, with no claim that ordinary trained networks satisfy it. The main learning statements and polynomial corollaries have Lean proofs.

1 The learning question and the restriction

We study a Bayesian network trained on noiseless examples from a Boolean circuit. The learner may have arbitrarily large width, and both the teacher's width and depth are allowed to grow. Our goal is a sample bound polynomial in these teacher resources and independent of learner width. We obtain it by conditioning the Gaussian prior on a concrete property: averaging neurons within suitable populations must approximately preserve the network's clipped output on every Boolean input.

This merging property supplies a finite description with polynomially many coefficients. Product-of-layer-norm capacity estimates give fixed-depth ReLU bounds whose degree in teacher width grows with depth; counting the merged descriptions instead gives polynomial dependence on depth as well. The restriction is independent of the target. A positive-mass neighborhood of a compiled Boolean circuit satisfies it, so the restricted posterior remains defined after every noiseless training sample.

Two features make the construction work for these activations. ReLU's positive homogeneity removes population sizes from the normalized computation, while GELU requires additional size

*Authors are listed in alphabetical order.

parameters whose effect can be controlled uniformly. Although perturbations can amplify exponentially through depth, their admissible parameter radius enters the prior-mass estimate only logarithmically. The result concerns exact Bayesian conditioning. It supplies neither an efficient posterior sampler nor a description of the output law of gradient descent.

2 Model and population condition

Fix $d \geq 1$, $L \geq 2$, $H = L - 1$, and $\mathcal{X} = \{-1, 1\}^d$. Write $\bar{x} = (x, 1)$. All hidden layers have width N:

$$h^1 = \mathrm{ReLU}(W^1 \bar{x}), \quad h^s = \mathrm{ReLU}(W^s h^{s-1}) \ (2 \leq s \leq H), \quad z_w = v^\top h^H.$$

The appended input coordinate supplies first-layer offsets. There are no separate internal biases; a constant wire used for later offsets is an ordinary neuron counted in the layer width. The raw prior $\mu_{N,t}$ has independent Gaussian entries of variances

$$t^2/d \text{ in } W^1, \qquad t^2/N \text{ in } W^s \ (s \geq 2), \qquad t^2/N^2 \text{ in } v, \quad 0 < t \leq 1.$$

The value $t = 1$ is the mean-field scale; smaller t gives a stronger prior. Thus $t = N^{(1-\gamma)/2}$ includes every $\gamma \geq 1$. Set

$$E_v = N \|v\|_2^2, \quad M(w) = \max_x |z_w(x)|, \quad T(u) = \min\{1/2, \max\{-1/2, u\}\}.$$

At layer s, partition the neurons into q_s nonempty populations, of arbitrary sizes $k_{s,a}$. Let column a of F_s be the normalized indicator of its population, with entries $1/\sqrt{k_{s,a}}$. Thus $F_s^\top F_s = I$. Put $P_s = F_s F_s^\top$ and define merged parameters

$$\bar{W}^1 = P_1 W^1, \quad \bar{W}^s = P_s W^s P_{s-1}, \quad \bar{v} = P_H v.$$

With $q = \max_s q_s$, the event $\mathcal{A}_N$ requires some such partitions satisfying

$$(q-1) M(w)^2 \leq 4 E_v(w), \qquad \max_x |T(z_w(x)) - T(z_{\bar{w}}(x))| \leq 1/16. \tag{1}$$

The condition does not use the target or its complexity. The count budget is invariant under readout rescaling; the clipped comparison retains fixed margins. The complete event is invariant under neuron permutations. Populations are unsigned: arbitrary hidden sign flips are not symmetries of ReLU networks. Finitely many witnesses at fixed N make the event Borel.

The prior is $\mu_{N,t}$ conditioned on $\mathcal{A}_N$. One may additionally impose the all-input margin event $|z_w(x)| > 1/2$ everywhere. Training further conditions on $f(X_i) z_w(X_i) > 1/2$. The theorem holds for either prior convention and tests the event $f(X) z_w(X) \leq 1/4$. Thus the training margin is larger than the test margin; the gap allows merging and coefficient rounding while retaining a useful prediction guarantee.

3 Explicit learning theorem

The target is a single-output layered Boolean circuit of width $C \geq 1$ and H stages, using NOR, COPY, and NOT. One constant-one wire is carried through every stage, including the final stage, and counts toward C. Compilation from another bounded-fan-in basis must count its width and depth overhead. Bits are $0, 1$; target labels are $-1, 1$.

The constant wire supplies internal offsets and the final readout's conversion from bits to signs. If an ordinary width-C_0 circuit has no reserved constant wire, adding one at each stage gives the

theorem with $C = C_0 + 1$, hence learner width $N \geq C_0 + 1$ for ReLU and $N \geq 2(C_0 + 1)$ for the GELU compilation below. The overhead matters at small widths: a width-one ReLU network with a bias-free linear readout has output of one weak sign everywhere, so it cannot learn a nonconstant Boolean literal with strict signed margins.

The first group of constants bounds the negative logarithm of the raw probability of a neighborhood implementing the target. The radius may shrink exponentially with depth, but its logarithm is the quantity used in the mass bound. Define

$$\begin{aligned}
&a = 4, && b_L = \sqrt{16\log 9 + 8\log(4L)}, && S_0 = 2 + 2\sqrt{10C} + b_L,\\
&D_0 = d + 1 + H\sqrt{C}, && \rho = \min\left\{\frac{1}{16}, \frac{1}{64a\sqrt{10C}\, D_0 S_0^H}\right\},\\
&q_0 = d + 2 + (H-1)C, && B_0 = d + 10(H-1)C + 10a^2 C,\\
&D_* = B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2. &&
\end{aligned} \tag{2}$$

The next budgets bound readout energy, the number of populations, and the coefficients of a merged network. Rounding those coefficients produces a finite set of predictions whose logarithmic size determines the sample bound. For $0 < \varepsilon, \delta < 1$, set

$$\begin{aligned}
&B = \max\{8, 8[D_* + \log(4/\varepsilon)]\}, && Q = \lceil 1 + 16B \rceil,\\
&J = 4\left[D_* + \log(4/\varepsilon) + 2H\log Q + \frac{\log 2}{2}\{d + 2 + (H-1)Q\}\right], &&\\
&R = 1 + \sqrt{J}, && A_g = (d + 1 + Q)(R + 1),\\
&\zeta = [32(H+1)A_g^{H+1}]^{-1}, && P_Q = Q(d+2) + (H-1)Q^2.
\end{aligned} \tag{3}$$

Theorem 1. *If*

$$n \geq \frac{2}{\varepsilon}\left[P_Q \log(3 + 4R/\zeta) + \log(1/\delta)\right], \tag{4}$$

then, with probability at least $1 - \delta$ over iid inputs from any fixed distribution on $\mathcal{X}$, the posterior obeys

$$\mathbb{E}_w \mathbb{P}_X\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ and $0 < t \leq 1$. All prior and posterior normalizers are positive. The same conclusion holds with the additional global-margin restriction.

At fixed accuracy and confidence, a useful summary is

$$n = \widetilde{O}(LdQ + L^2Q^2), \qquad Q = \widetilde{O}(Ld + L^2 C).$$

More generally the leading $1/\varepsilon$ and confidence term are explicit, and Q additionally depends logarithmically on $1/\varepsilon$. All depth dependence is polynomial. At fixed other parameters the width dependence is $O(C^2 \log^3(2 + C))$.

Corollary 2. *For a universal constant A and $S = 2 + d + L + \log(4/\varepsilon)$, a sufficient bound for these single-output bounded-fan-in teachers is*

$$n \geq \frac{A}{\varepsilon}\left[C^2 S^9 + \log(1/\delta)\right].$$

This coarser estimate prunes irrelevant circuit wires before absorbing logarithmic width terms into depth. Neither estimate is a claim of optimality.

4 A positive-mass Boolean implementation

The lemma below makes the learning model nonempty at every allowed width and scale. Its target neighborhood lies in the merging event and has enough mass to lower-bound every noiseless posterior denominator.

Lemma 3 (Target cylinder)**.** *For every $N \geq C$ and $0 < t \leq 1$, a set of raw probability at least $\exp(-ND_*/t^2)$ lies in $\mathcal{A}_N$ and satisfies $f(x)z_w(x) > 1$ everywhere. It also satisfies the optional global margin.*

Proof. Partition the neurons into C balanced clone groups of sizes k_b, so $C/2 \leq N/k_b \leq 2C$. Implement gates by

$$\mathrm{NOR}(u, v) = \mathrm{ReLU}(1 - u - v), \quad \mathrm{COPY}(u) = \mathrm{ReLU}(u), \quad \mathrm{NOT}(u) = \mathrm{ReLU}(1 - u).$$

Propagate a constant-one wire by a copy. Substitute input bits $(1 + x_i)/2$ into first-layer gates. Each first-layer row has squared norm at most one. Each later logical row has sum of squared coefficients at most five, including repeated gate inputs. Divide each source coefficient equally among that source's clones. The teacher states $b^s(x)$ are exact $0, 1$ bits, constant on populations, with norm at most $\sqrt{N}$. Read out $z_* = a(2b^H_{\mathrm{out}} - 1) = af$, using the final constant wire. Combine coefficients if the output and constant wires coincide.

Let P average within these fixed clone groups. The teacher satisfies

$$PW^1_* = W^1_*, \quad PW^s_* P = W^s_*, \quad Pv_* = v_*.$$

The coefficient bounds and clone sizes give

$$\left\|W^1_*\right\|^2_F \leq N, \quad \|W^s_*\|^2_F \leq 10C, \quad a^2C/2 \leq N \|v_*\|^2_2 \leq 10a^2C.$$

The lower readout bound remains valid when output and constant wires coincide.

In raw coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv, constrain each first-layer coordinate, each internal row's coordinate along every normalized clone-group direction, and each readout coordinate to an interval of radius ρt around its teacher value. There are Nq_0 independent constrained coordinates, with sum of squared centers at most NB_0. Internally

$$W^s = W^s_* + R^s + E^s, \quad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t\sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk is independent of these coordinates. Two $1/4$-nets of size at most 9^N give

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2\exp(2N \log 9 - Nb^2_L/8) \leq 1/(2L).$$

Thus all bulk bounds hold conditionally with probability at least $1/2$.

Set $e_s = \|h^s - b^s\|_2 / \sqrt{N}$. ReLU is 1-Lipschitz, giving $e_1 \leq \rho(d + 1)$. Since $E^s b^{s-1} = 0$ and $\|b^{s-1}\|_2 / \sqrt{N} \leq 1$,

$$e_s \leq (\sqrt{10C} + \rho\sqrt{C} + b_L)e_{s-1} + \rho\sqrt{C} \leq S_0 e_{s-1} + \rho\sqrt{C}.$$

Hence $e_s \leq \rho D_0 S_0^{s-1}$ and $e_H \leq 1/(64a\sqrt{10C})$. Also $\sqrt{N}\|v - v_*\|_2 \leq \rho$, so

$$|z_w - z_*| \leq a\sqrt{10C}\, e_H + \rho(1 + e_H) < 1/8.$$

This proves signed margin greater than one and $M(w) \leq a + 1/8 \leq 5a/4$.

The clone projections map this cylinder into itself. Row averaging preserves coordinate intervals because teacher rows agree within populations; right projection kills the bulk. The first-layer and readout intervals are likewise preserved. The merged network has the same strict signed margin, so both clipped outputs equal $f/2$. Moreover

$$\sqrt{E_v(w)} \geq a\sqrt{C/2} - \rho \geq (5/8)a\sqrt{C},$$

which gives $(C-1)M(w)^2 \leq 4E_v(w)$. Thus the cylinder lies in $\mathcal{A}_N$.

For $Z \sim N(0, t^2)$ and any center m, direct integration of the Gaussian density gives

$$\mathbb{P}\{|Z - m| \leq \rho t\} \geq \frac{2\rho}{e\sqrt{2\pi}} e^{-m^2/t^2} \qquad (\rho \leq 1/16).$$

Multiply over the constrained coordinates and include the bulk probability $1/2$. Since $t \leq 1$ and $N/t^2 \geq 1$, the result is at least e^{-ND_*/t^2}. □

The radius decreases exponentially with depth. Only its logarithm enters the mass budget. The N^2 Gaussian bulk coordinates are never forced into a small Euclidean ball.

5 Compressed coordinates and finite coding

For each fixed choice of populations, we can express the merged network using one coordinate per population. A Gaussian tail bound then controls these coordinates even when the witnessing populations are selected after the network is drawn. The finite grid obtained by rounding is used only to compare predictions and count possibilities; it is an auxiliary class for the proof. For fixed populations put

$$B^1 = F_1^\top W^1/\sqrt{N}, \quad B^s = F_s^\top W^s F_{s-1}, \quad b = \sqrt{N} F_H^\top v.$$

The merged network is exactly

$$g^1 = \mathrm{ReLU}(B^1 \bar{x}), \quad g^s = \mathrm{ReLU}(B^s g^{s-1}), \quad z_{\bar{w}} = b^\top g^H.$$

Its full hidden state is $\sqrt{N} F_s g^s$; disjoint nonnegative columns and positive homogeneity let ReLU commute with this lift. No group-size coordinates are needed.

The entries of $\sqrt{d}B^1$, the internal B^s, and b are independent Gaussians of variance t^2/N. Set

$$\kappa = d\left\|B^1\right\|_F^2 + \sum_{s=2}^{H} \|B^s\|_F^2 + \|b\|_2^2, \quad p = q_1(d+1) + \sum_{s=2}^{H} q_s q_{s-1} + q_H.$$

Then

$$\mathbb{E}e^{N\kappa/(4t^2)} = 2^{p/2}, \qquad p/N \leq d + 2 + (H-1)Q \quad (q_s \leq \min\{N, Q\}). \tag{5}$$

Proof of Theorem 1. The target cylinder supplies a lower bound e^{-ND_*/t^2} on every noiseless posterior normalizer relative to the raw Gaussian, with either support convention. Since NE_v/t^2 is chi-square with N degrees of freedom,

$$\mu_{N,t}(E_v > B) \leq e^{-NB/(8t^2)} \qquad (B \geq 8).$$

Its posterior probability is at most $\varepsilon/4$. A nonempty training set forces $M > 1/2$, so (1) implies $q < 1 + 16E_v$. Hence witnessing population counts on the retained posterior are at most Q.

There are at most $Q^{H(N+1)}$ tuples of partitions with counts at most $\min\{N, Q\}$. By (5), the raw probability that any tuple has $\kappa > J$ is at most

$$\exp\left[H(N+1)\log Q + \frac{N}{2}\{d+2+(H-1)Q\}\log 2 - \frac{NJ}{4t^2}\right].$$

Using $t \le 1, N \ge 1$ and (3), this is at most $e^{-ND_*/t^2}(\varepsilon/4)^{N/t^2}$. Its posterior probability is at most $\varepsilon/4$. The union is taken before selecting a witness from the realized network.

Every remaining network has a merged representation with coefficients bounded by $\sqrt{J}$. Pad to width Q with zeros, and round coefficients to fixed grids with coordinate error at most ζ, preserving zero. Both original and rounded coefficients have magnitude at most R. Hidden-coordinate magnitudes are bounded by A_g^s. Rounding errors obey

$$e_1 \le A_g\zeta, \qquad e_s \le A_g e_{s-1} + \zeta A_g^s, \qquad e_s \le s\zeta A_g^s.$$

The readout error is at most $(H+1)\zeta A_g^{H+1} = 1/32$. Thus the grid class $\mathcal{G}$, independent of N, t, satisfies

$$\log|\mathcal{G}| \le P_Q \log(3 + 4R/\zeta), \qquad |T(z_w) - T(g)| \le 3/32.$$

The second bound combines merging and rounding. Training inputs have $fT(g) \ge 13/32$, while a tested failure $fz_w \le 1/4$ entails $fT(g) \le 11/32$.

For fixed g, if the set $\{x : f(x)T(g(x)) \le 11/32\}$ has probability greater than $\varepsilon/2$, a sample avoiding it has probability at most $e^{-n\varepsilon/2}$. The union bound and (4) give one event on which every retained network has failure probability at most $\varepsilon/2$. The discarded posterior mass contributes at most $\varepsilon/2$. The common grid class proves simultaneity over all widths and scales. □

Proof of Corollary 2. Prune wires not ancestral to the single output, retaining required constants. Fan-in two yields width $C' \le \min\{C, 2^H + 1\}$ at the same depth. Apply the theorem with C' and use $\log(2 + C') = O(L)$. For $S = 2 + d + L + \log(4/\varepsilon)$, the explicit formulas give, with universal constants changing between inequalities,

$$\log(1/\rho) \le AS^2, \quad D_* \le AC'S^3, \quad Q \le AC'S^3, \quad J \le AC'S^4.$$

Hence $P_Q \le A(C')^2S^7$. Since $R \le A\sqrt{C'}S^2$ and $A_g \le A(C')^{3/2}S^5$, pruning also gives

$$\log(3 + 4R/\zeta) \le AS^2.$$

Thus $\log|\mathcal{G}| \le A(C')^2S^9 \le AC^2S^9$, proving the bound. □

6 The same conclusion for GELU

Let $\phi(u) = u\Phi(u)$ be exact GELU, where Φ is the standard normal distribution function. Replace every hidden ReLU by ϕ, and define the prior event (1) using this network and its merged counterpart. The same structural condition gives a polynomial-depth theorem. Here the Boolean circuit's compilation to GELU has a counted factor-two width overhead.

For a width-C Boolean teacher let $D = 2C$, and replace the target-cylinder budgets (2) by

$$S_{\mathrm{G}} = 4 + 4\sqrt{100D} + 2b_L, \qquad D_{\mathrm{G},0} = 2(d+1) + 2H\sqrt{D},$$

$$\rho_{\mathrm{G}} = \min\left\{\frac{1}{16}, \frac{1}{64a\sqrt{100D}\, D_{\mathrm{G},0} S_{\mathrm{G}}^H}\right\}, \qquad q_{\mathrm{G},0} = d + 2 + (H-1)D,$$

$$B_{\mathrm{G},0} = d + 100(H-1)D + 100a^2D, \qquad D_{\mathrm{G},*} = B_{\mathrm{G},0} + q_{\mathrm{G},0}\log\frac{e\sqrt{2\pi}}{2\rho_{\mathrm{G}}} + \log 2.$$

Use $D_{\mathrm{G},*}$ in place of D_* in the definitions of B, Q, J, R. Use instead

$$A_{\mathrm{G}} = 4(d+1+Q)(R+1), \qquad \zeta_{\mathrm{G}} = [32(H+1)A_{\mathrm{G}}^{H+1}]^{-1}, \qquad P_{\mathrm{G}} = P_Q + HQ.$$

Theorem 4 (GELU version)**.** *Theorem 1 holds for the GELU prior just defined, simultaneously for every $N \geq 2C$ and $0 < t \leq 1$, if its sample bound is replaced by*

$$n \geq \frac{2}{\varepsilon}\left[P_{\mathrm{G}} \log(3 + 4R/\zeta_{\mathrm{G}}) + \log(1/\delta)\right].$$

The optional all-input margin restriction is still allowed. The width dependence remains $O(C^2 \log^3(2+C))$ at fixed other parameters. Corollary 2 holds after changing its universal constant.

Proof. We give the two changes needed to the ReLU proof: an exact Boolean compiler and a uniformly well-conditioned family of compressed activations.

Exact Boolean compiler. Set $\tau = \Phi(1) > 2/3$, $s = 2\tau - 1$, $A = \tau/s$, and $B = (1-\tau)/s$. Then $1 < A < 2$ and $0 < B < 1$, and direct substitution gives

$$A\phi(u) + B\phi(-u) = \mathrm{ReLU}(u) \qquad (u \in \{-1, 0, 1\}).$$

Represent every logical bit, including every constant-one wire, by two GELU neurons with preactivations $u, -u$. Decode its bit by the displayed linear combination. For NOR, COPY, and NOT use respectively $u = 1 - b_1 - b_2$, $u = b_1$, and $u = 1 - b_1$. Decode the constant-one wire for the internal offset. First-layer input bits are $(1 + x_i)/2$. All preactivations lie in $\{-1, 0, 1\}$ on Boolean inputs, so the simulation is exact at the same depth and width $D = 2C$.

Every hidden coordinate has magnitude at most one. First-layer row squared norms are at most one. An internal row has squared norm at most $5(A^2 + B^2) < 25$, including repeated gate inputs. Before cloning, the readout for $af = a(2b_{\mathrm{out}} - 1)$ has squared coefficient norm between a^2 and $25a^2$: if output and constant coincide, combine their coefficients. Balanced cloning into width $N \geq D$ therefore supplies the conservative bounds

$$\|W_*^1\|_F^2 \leq N, \qquad \|W_*^s\|_F^2 \leq 100D, \qquad a^2 D/2 \leq N\|v_*\|_2^2 \leq 100a^2 D.$$

These exact teacher states are constant on the D clone populations.

GELU is globally 2-Lipschitz and vanishes at zero. The cylinder proof therefore has error recurrences

$$e_1 \leq 2\rho_{\mathrm{G}}(d+1), \qquad e_s \leq S_{\mathrm{G}} e_{s-1} + 2\rho_{\mathrm{G}}\sqrt{D}.$$

The chosen radius gives $e_H \leq 1/(64a\sqrt{100D})$. The same readout and projection calculation proves signed output margin greater than one, clipped merging equality, and the normalized population-count budget. Its raw probability is at least $e^{-ND_{\mathrm{G},*}/t^2}$. Although the approximate network is not an exact digital circuit, its errors are controlled by the chosen parameter radius; digital restoration is not needed.

Population masses in the compressed model. For a population of size k, put $r = \sqrt{k/N} \in (0, 1]$ and define

$$\psi_r(u) = r\phi(u/r), \qquad \psi_0(u) = \mathrm{ReLU}(u).$$

Writing φ for the standard normal density, differentiation for $r > 0$ gives

$$\partial_r \psi_r(u) = \phi(u/r) - (u/r)\phi'(u/r) = -(u/r)^2 \varphi(u/r), \qquad |\partial_r \psi_r(u)| \leq \frac{2}{e\sqrt{2\pi}} < 1.$$

Also $|\phi'| \le 2$. The extension at $r = 0$ is continuous uniformly in u, since $\sup_u |\phi(u) - \mathrm{ReLU}(u)| < \infty$. Consequently

$$|\psi_r(u) - \psi_{r'}(u')| \le 2|u - u'| + |r - r'|, \qquad |\psi_r(u)| \le |u| \quad (0 \le r, r' \le 1). \tag{6}$$

The exact normalized merged computation is now

$$g_j^1 = \psi_{r_{1,j}}((B^1 \bar{x})_j), \qquad g_j^s = \psi_{r_{s,j}}((B^s g^{s-1})_j), \qquad z = b^\top g^H, \quad r_{s,j} = \sqrt{k_{s,j}/N}.$$

The Gaussian distribution of $\sqrt{d}B^1, B^s, b$ is unchanged. Thus the same readout and partition-union bounds retain a posterior of mass at least $1 - \varepsilon/2$ with $q_s \le Q$ and coefficient norm budget J.

For finite coding, append at most HQ population-mass parameters in $[0, 1]$, including zero masses for padded zero neurons. Round these and all coefficients with coordinate error at most ζ_G. Equation (6) and the enlarged A_G give hidden magnitudes at most A_G^s and rounding errors at most $s\zeta_G A_G^s$. The output error is at most $1/32$. The grid has log-cardinality at most $P_G \log(3 + 4R/\zeta_G)$. Its elements are auxiliary real-valued predictors; they need not themselves be width-N networks or have realizable population proportions. The finite-class argument uses only their predictions and therefore applies unchanged. The same ancestral pruning and estimates as in Corollary 2 give the stated literal polynomial. □

7 Interpretation and limits

If $p(x) = \mathbb{P}_w\{f(x)z_w(x) > 1/4\}$, then the theorem implies

$$\mathbb{P}_X\{p(X) \le \theta\} \le \varepsilon/(1 - \theta), \qquad 0 < \theta < 1.$$

The strict $> 2/3$ criterion follows by adjusting accuracy by a factor of three. Correct-sign probability is at least the correct-margin probability. These conclusions concern the full posterior distribution, including mixtures.

The restriction expresses stability under a specific structured change of the weights. It is not proved equivalent to robustness against arbitrary calibrated internal disturbances, nor established for typical trained networks. Its positive-mass Boolean implementation shows useful representability, not universal empirical applicability. An arbitrary analog ReLU teacher is not identified with a Boolean circuit of the same width and depth.

These theorems therefore give polynomial-depth results for specified restricted priors, at all mean-field-or-stronger scales. They do not settle the unrestricted polynomial-depth learning question. Exponentially small parameter radii are harmless in this construction because only their logarithm is paid; exponentially large compressed width would not be harmless, and the population condition prevents that loss.

Provenance and verification. The population construction and Gaussian partition estimates build on collaborative work on tanh networks, including contributions from Claude. This paper supplies the ReLU normalized representation, GELU perspective extension and exact compiler, noncontracting target-cylinder estimate, and resulting depth-polynomial theorems. The main learning statements have Lean proofs, with their scope recorded in the repository.